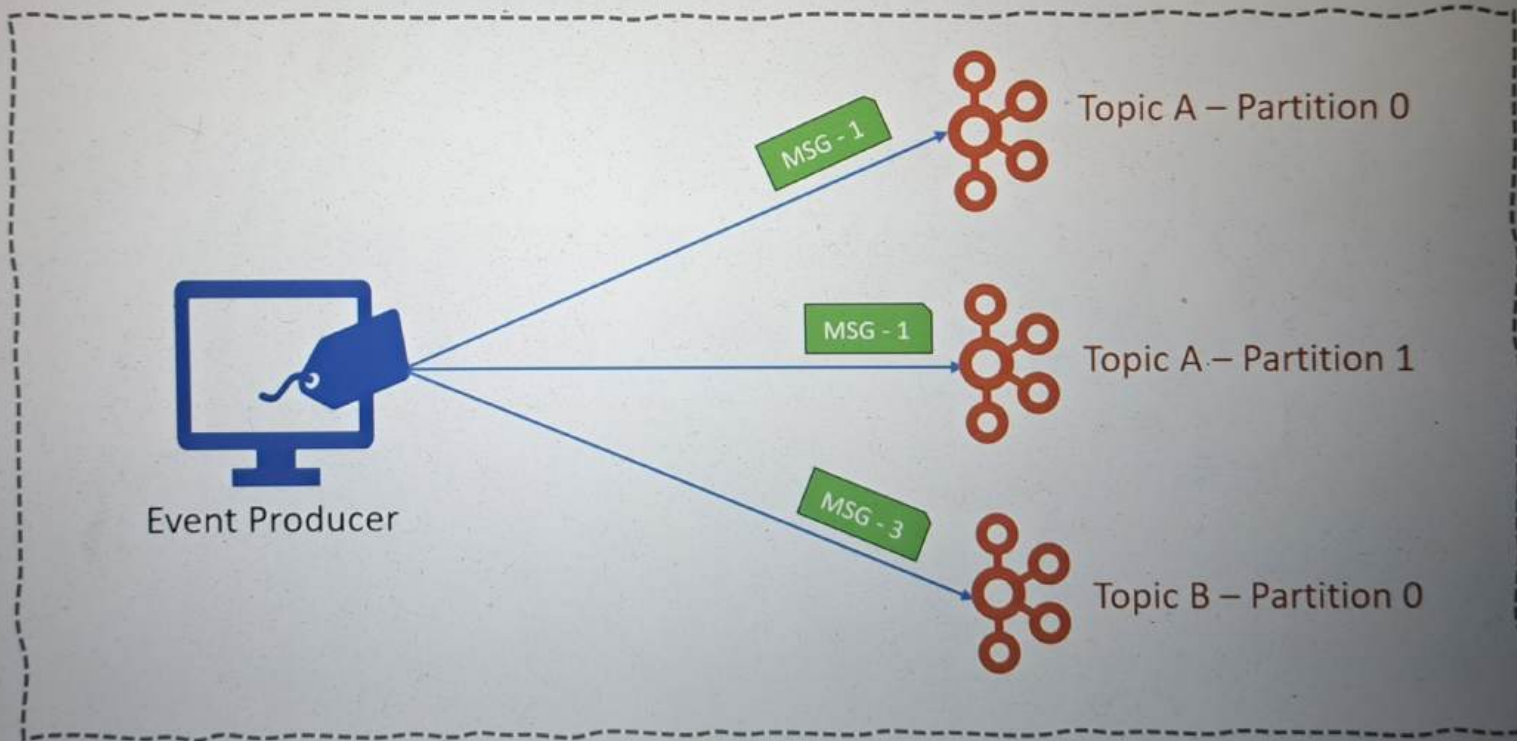


Transactions in Kafka Producer?

Implementing Transactions



Single Transaction – All or Nothing

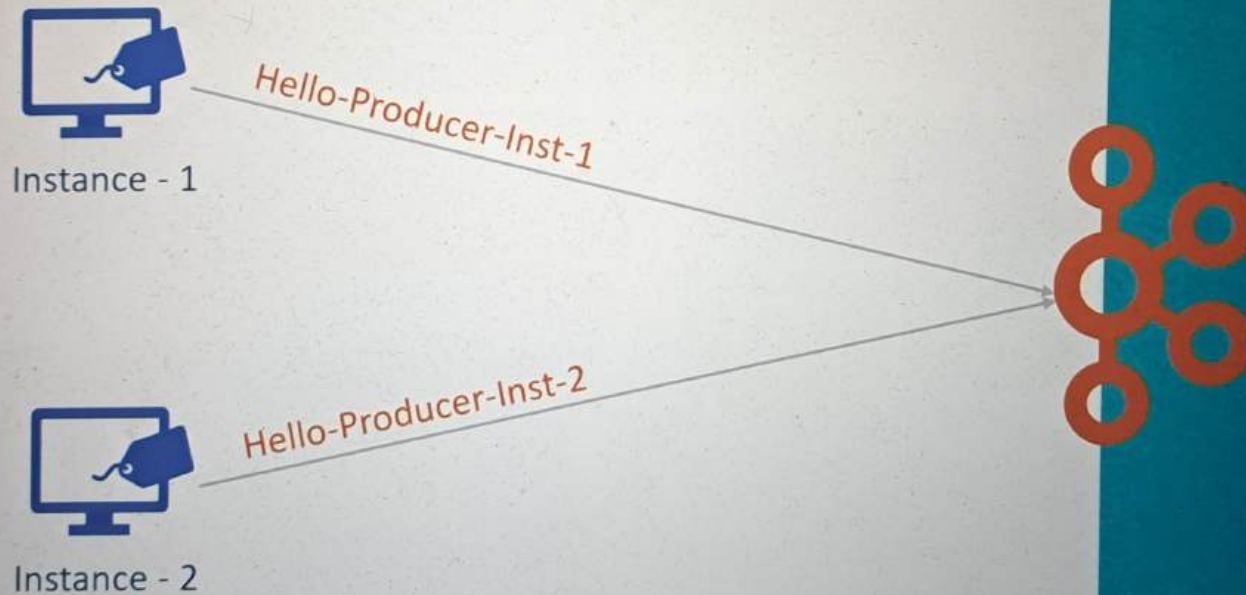
Implementing Transactions

1. Transaction depends on Idempotence
- ➔ 2. transactional_id_config must be unique

Implementing Transactions



Implementing Transactions



```
20 props.put(ProducerConfig.CLIENT_ID_CONFIG, AppConfigs.applicationID);
21 props.put(ProducerConfig.BOOTSTRAP_SERVERS_CONFIG, AppConfigs.bootstrapServers);
22 props.put(ProducerConfig.KEY_SERIALIZER_CLASS_CONFIG, IntegerSerializer.class.getName());
23 props.put(ProducerConfig.VALUE_SERIALIZER_CLASS_CONFIG, StringSerializer.class.getName());
24 props.put(ProducerConfig.TRANSACTIONAL_ID_CONFIG, AppConfigs.transaction_id);
25
26 KafkaProducer<Integer, String> producer = new KafkaProducer<>(props);
27 producer.initTransactions();
28
29 logger.info(s: "Starting First Transaction...");
30 producer.beginTransaction();
31 try {
32     for (int i = 1; i <= AppConfigs.numEvents; i++) {
33         producer.send(new ProducerRecord<>(AppConfigs.topicName1, i, value: "Simple Message-T1-" + i));
34         producer.send(new ProducerRecord<>(AppConfigs.topicName2, i, value: "Simple Message-T1-" + i));
35     }
36     logger.info(s: "Committing First Transaction.");
37     producer.commitTransaction();
38 } catch (Exception e) {
39     logger.error(s: "Exception in First Transaction. Aborting...");
40     producer.abortTransaction();
41     producer.close();
42     throw new RuntimeException(e);
43 }
44
45 logger.info(s: "Finished - Closing Kafka Producer.");
46 producer.close();
47
48 }
49
50 }
```

transactional-producer [C:\Udemy\StarterProject\04-transactional-producer] - ...src\main\java\guru\learningjournal\kafka\examples\HelloProducer.java [transactional-producer] - IntelliJ IDEA

File Edit View Navigate Code Analyze Refactor Build Run Tools VCS Window Help

04-transactional-producer src main java guru learningjournal kafka examples HelloProducer

Project

- 04-transactional-producer [transactional-producer]
- idea
- scripts
 - 0-kafka-server-start.cmd
 - 1-kafka-server-start.cmd
 - 2-kafka-server-start.cmd
 - consumer-start.cmd
 - topic-1-create.cmd
 - topic-2-create.cmd
 - zookeeper-start.cmd
- src
 - main
 - guru.learningjournal.kafka.examples
 - AppConfigs
 - HelloProducer
 - resources
 - log4j2.xml
 - test
 - target
 - tmp
 - pot.xml
 - transactional-producer.iml
 - External Libraries
 - Scratches and Consoles

AppConfigs.java

HelloProducer.java

```
logger.info("Starting First Transaction...");
producer.beginTransaction();
try {
    for (int i = 1; i <= AppConfigs.numEvents; i++) {
        producer.send(new ProducerRecord<>(AppConfigs.topicName1, i, value: "Simple Message-T1-" + i));
        producer.send(new ProducerRecord<>(AppConfigs.topicName2, i, value: "Simple Message-T1-" + i));
    }
    logger.info("Committing First Transaction.");
    producer.commitTransaction();
} catch (Exception e) {
    logger.error("Exception in First Transaction. Aborting...");
    producer.abortTransaction();
    producer.close();
    throw new RuntimeException(e);
}

logger.info("Starting Second Transaction...");
producer.beginTransaction();
try {
    for (int i = 1; i <= AppConfigs.numEvents; i++) {
        producer.send(new ProducerRecord<>(AppConfigs.topicName1, i, value: "Simple Message-T2-" + i));
        producer.send(new ProducerRecord<>(AppConfigs.topicName2, i, value: "Simple Message-T2-" + i));
    }
    logger.info("Aborting Second Transaction.");
    producer.abortTransaction();
} catch (Exception e) {
    logger.error("Exception in Second Transaction. Aborting...");
    producer.abortTransaction();
    producer.close();
    throw new RuntimeException(e);
}

logger.info("Finished - Closing Kafka Producer.");
```

HelloProducer main()

Run TODO Terminal

All files are up-to-date (3 minutes ago)

52:41 CRLF: UTF-8: 4spaces

Implementing Transactions